

ximera — Simultaneously write print and online interactive materials.*

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Abstract

“Ximera begins where $\text{\TeX}$ ends.” The `ximera` class aids in the creation of hand-outs, worksheets, exercises, and sections of textbooks to be used either individually or “glued” together via a `xourse` file. All `ximera` documents can be deployed in an online interactive form via `xake`. See: [Ximera Project](#) and the source code on [GitHub](#).

1 Introduction

Ximera, pronounced “chimera,” (**X**imera: **I**nteractive, **M**athematics, **E**Resources, for **A**ll) is an open-source platform that provides tools for authoring and publishing (PDF and Online), open-source, interactive educational content, such as textbooks, assessments, and online courses. The Ximera document class provides the following features:

Formatting for different domains The Ximera document class provides built-in support for formatting documents in both PDF and online formats, which can be a big time-saver for authors. Additionally, it allows for the simultaneous creation of solution manuals and teaching editions, which can be especially useful for educators.

Compiling individually or as a whole With the Ximera document class, authors can easily compile individual documents or an entire collection of documents. This flexibility can be helpful when making changes to specific documents without having to re-compile the entire collection. Moreover, this allows an author to share large portions of a text with another, with minimal changes.

Interactive content The Ximera document class allows for the inclusion of interactive content, such as answer boxes that are validated by a client-side computer algebra system. Additionally, it allows for the embedding of YouTube videos, Desmos graphs, and GeoGebra interactives.

All content displayed By default, the Ximera document class displays all content to the author. This means the author sees what the students see, along with answers and solutions, and links (that can be checked) to various interactive elements (when deployed, the interactive elements are truly embedded). This can be especially helpful for catching errors or inconsistencies in the content.

Online examples can be found at

<https://go.osu.edu/ximera-examples>

*This file describes version v1.5.1, last revised 2024/05/12.

2 ximera.cls

```
1 \classXimera
2 \newif\ifnumberedProblems
3 \numberedProblemsfalse% Default to no numbers, as that was previous behavior.
4 \DeclareOption{onlineProblemNumbers}{\numberedProblemstrue}
5 \endclassXimera
```

2.1 Options for the class

We start by listing the options for the ximera document class. Note, since the xourse class is based on the ximera class, all listed options are available there too.

```
6 \classXimera
```

The default behavior of the class is to display **all** content. This means that if any questions are asked, all answers are shown. Moreover, some content will only have a meaningful presentation when displayed online. When compiled without any options, this content will be shown too. This option will suppress such content and generate a reasonable printable “handout.”

```
7 \newif\ifhandout
8 \handoutfalse
9 \DeclareOption{handout}{\handouttrue}
```

By default, authors are listed at the bottom of the first page of a document. This option will suppress the listing of the authors.

```
10 \newif\ifnoauthor
11 \noauthorfalse
12 \DeclareOption{noauthor}{\noauthortrue}
```

By default, learning outcomes are listed at the bottom of the first page of a document. This option will suppress the listing of the learning outcomes.

```
13 \newif\ifnooutcomes
14 \nooutcomesfalse
15 \DeclareOption{nooutcomes}{\nooutcomestru}
```

instructornotes This option will turn on (and off) notes written for the instructor.

```
16 \newif\ifinstructornotes
17 \instructornotesfalse
18 \DeclareOption{instructornotes}{\instructornotestru}
```

noinstructornotes This option will turn off (and on) notes written for the instructor.

```
19 \DeclareOption{noinstructornotes}{\instructornotestru}
```

hints When the `handout` options is used, hints are not shown. This option will make hints visible in handout mode.

```
20 \newif\ifhints
21 \hintsfalse
22 \DeclareOption{hints}{\hintstrue}
```

newpage This option will start each problem-like environment (`exercise`, `question`, `problem`, and `exploration`) start on a new page.

```
23 \newif\ifnewpage
24 \newpagefalse
25 \DeclareOption{newpage}{\newpagetrue}
```

numbers This option will number the titles of the activity. By default the activities are unnumbered.

```
26 \newif\ifnumbers
27 \numbersfalse
28 \DeclareOption{numbers}{\numberstrue}
```

`wordchoicegiven` This option will replace the choices shown by `wordChoice` with the correct choice. No indication of the `wordChoice` environment will be shown.

```

29 \newif\ifwordchoicegiven
30 \wordchoicegivenfalse
31 \DeclareOption{wordchoicegiven}{\wordchoicegiventrue}
32 \newif\iffirstinlinechoice% Support for other wordchoice command contents.
33 \firstinlinechoicetrue

34
35 \newif\ifxake
36 \xakefalse
37 \DeclareOption{xake}{\xaketrue}
38
39 \newif\iftikzexport
40 \tikzexportfalse
41 \DeclareOption{tikzexport}{%
42   \tikzexporttrue%
43   \handoutfalse%
44   \numbersfalse%
45   \newpagefalse%
46   \hintsfalse%
47   \nooutcomesfalse%
48 }
49
50 \DeclareOption*{%
51   \PassOptionsToClass{\CurrentOption}{article}%
52 }
53 \ProcessOptions\relax
54 \LoadClass{article}
55
56 \ifdefined\HCode
57   \xaketrue%
58   \tikzexporttrue%
59   \handoutfalse%
60   \numbersfalse%
61   \newpagefalse%
62   \hintsfalse%
63   \nooutcomesfalse%
64 \fi

65 </classXimera>
66 <*classXimera>

```

2.2 Loading packages

Since we want `\cancel` to work, we load it here to avoid polluting the `.jax` output.

```
67 \RequirePackage[makeroom]{cancel}
```

Quite a few packages are required by the document class. This is a list of required packages. As packages are added to this list, we should include a comment as to where they are being utilized. This will help keep this list from being redundant and/or outdated.

```

68 \RequirePackage[inline]{enumitem}
69 \RequirePackage[pagestyles]{titlesec}
70 \RequirePackage{titletoc}
71 \RequirePackage{titling}
72 \RequirePackage{url}
73 \RequirePackage[table]{xcolor}
74 \RequirePackage{tikz}
75 \RequirePackage{pgfplots}
76 \usepgfplotslibrary{groupplots}
77 \usetikzlibrary{calc}
78 \RequirePackage{fancyvrb}

```

Load `forloop` for the problem environment dynamic naming and building.

```
79 \RequirePackage{forloop}
```

Now we load even more packages.

```
80 \RequirePackage{environ}% Included to allow saving of environment contents. This does *not* p
81 \RequirePackage{amssymb}% Included to have access to math typeset.
82 \RequirePackage{amsmath}% Included to have access to math typeset.
83 \RequirePackage{amsthm}% Included to have access to math typeset.
84 \RequirePackage{xifthen}% http://ctan.org/pkg/xifthen
85 \RequirePackage{multido}% http://ctan.org/pkg/multido
86 \RequirePackage{listings} %% is this required???
87
88 \RequirePackage{xkeyval}
89
90 \RequirePackage{currfile}
91 \RequirePackage{comment}
92 \end{classXimera}
```

Various packages must be loaded early to avoid polluting the `.jax` file.

```
93 \begin{classXimera}
94 \RequirePackage{getttitlestring}
95 \RequirePackage{nameref}
96 \RequirePackage{epstopdf}
97 \end{classXimera}
```

2.3 Page setup

We want non-indented spaced-out paragraphs.

```
98 \begin{classXimera}
99 \setlength{\parindent}{0pt}
100 \setlength{\parskip}{5pt}
101 \end{classXimera}
```

To avoid weird margins in 2-sided mode, change the margins.

```
102 \begin{classXimera}
103 \oddsidemargin 62pt
104 \evensidemargin 62pt
105 \textwidth 345pt
106 \headheight 14pt
107 \end{classXimera}
```

On the HTML side, there is more complicated page setup to perform.

```
108 \begin{cfgXimera}
109 \Preamble{xhtml,mathjax,minipage-width}
110
111 % We don't want to translate font suggestions with ugly wrappers like
112 % <span class="cmti-10"> for italic text
113 \NoFonts
114
115 % Don't output xml version tag
116 % \Configure{VERSION}{}
117
118 % Output HTML5 doctype instead of the default for HTML4
119 % \Configure{DOCTYPE}{\HCode{<!doctype html>\Hnewline}}
120
121 % Custom page opening
122 % \Configure{HTML}{\HCode{<html lang="en">\Hnewline}}{\HCode{\Hnewline</html>}}
123
124 % Reset <head>, aka delete all default boilerplate; alternatively set up new content
125 % \Configure{@HEAD}{\HCode{<meta name="generator" content="TeX4ht (http://www.cse.ohio-state
126 \Configure{@HEAD}{\HCode{<meta name="ximera" content="version 2.5.1" />\Hnewline}}
127 \Configure{@HEAD}{\HCode{<link href="https://ximera.osu.edu/public/stylesheets/standalone.css
128 \Configure{@HEAD}{\HCode{<script type="text/javascript" async src="https://ximera.osu.edu/pub
129
```

```

130 % OVERWRITE css in ximera-server (to be removed whenever this has been fixed in the server;
131 \catcode'\%=11
132 \Configure{@BODY}{\HCode{<style>
133 .activity-body pre {
134     white-space: pre;
135     background-color: lightgray;
136 }
137 .xmyoutube {
138     aspect-ratio: 16/9;
139     min-width: 75%;
140 }
141 .image-environment img {
142     width: unset;
143 }
144 </style>\Hnewline}}
145 \catcode'\%=14
146
147 \</cfgXimera>

```

Disable certain ligatures in HTML.

```

148 \<htXimera>
149 \usepackage{microtype}
150 \DisableLigatures[f]{encoding=*}
151 \</htXimera>

```

I am not sure what this does.

```

152 \<htXimera>
153 \NewEnviron{html}{\HCode{\BODY}}
154 \</htXimera>

```

2.4 Structure

2.4.1 Macros

Makes everymath display style even when inline, could be optional.

```

155 \<classXimera>
156 \everymath{\displaystyle}
157 \</classXimera>

```

Ok not everything, we also need to configure “display style” limits.

```

158 \<classXimera>
159 \let\prelim\lim
160 \renewcommand{\lim}{\displaystyle\prelim}
161 \</classXimera>

```

2.4.2 Heading levels on the web

Machinery for emitting HTML headings whose level matches the surrounding document outline, so that heading-based navigation (screen readers) sees no skipped levels. `tex4ht` emits `\section` as `<h3>`, `\subsection` as `<h4>`, `\subsubsection` as `<h5>`; content directly under the (platform-rendered) activity title sits at level 3. `\xmheadlevel` computes the level for a heading emitted at the current position: 3 plus one per sectioning level already entered, plus `xmHeadOffset`. Any environment that emits its own heading and can contain further heading-emitting content (theorem-like environments, accordion items, expandables, hints) must step `xmHeadOffset` for the duration of its body so nested headings land one level deeper. Capped at 6 since HTML has no h7.

```

162 \<htXimera>
163 \newcounter{xmHeadOffset}
164 % Sectioning depth is tracked with hooks rather than by inspecting the
165 % section/subsection/subsubsection counters: ximera.cls sets secnumdepth to 2
166 % (or -2 in some modes), so those counters do not reliably step, and starred
167 % sectioning commands never step them. The hooks fire for starred forms too.
168 \newcounter{xmSecDepth}
169 \AddToHook{cmd/section/before}{\setcounter{xmSecDepth}{1}}

```


<code>conjecture (env.)</code>	Conjecture	
	213 <code>\classXimera</code>	<code>\newtheorem{conjecture}{\GetTranslation{Conjecture}}</code>
	214 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{conjecture}</code>
<code>corollary (env.)</code>	Corollary	
	215 <code>\classXimera</code>	<code>\newtheorem{corollary}{\GetTranslation{Corollary}}</code>
	216 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{corollary}</code>
<code>criterion (env.)</code>	Criterion	
	217 <code>\classXimera</code>	<code>\newtheorem{criterion}{\GetTranslation{Criterion}}</code>
	218 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{criterion}</code>
<code>definition (env.)</code>	Definition	
	219 <code>\classXimera</code>	<code>\newtheorem{definition}{\GetTranslation{Definition}}</code>
	220 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{definition}</code>
<code>example (env.)</code>	Example	
	221 <code>\classXimera</code>	<code>\newtheorem{example}{\GetTranslation{Example}}</code>
	222 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{example}</code>
<code>explanation (env.)</code>	Explanation	
	223 <code>\classXimera</code>	<code>\newtheorem*{explanation}{\GetTranslation{Explanation}}</code>
	224 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{explanation}</code>
<code>fact (env.)</code>	Fact	
	225 <code>\classXimera</code>	<code>\newtheorem{fact}{\GetTranslation{Fact}}</code>
	226 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{fact}</code>
<code>lemma (env.)</code>	Lemma	
	227 <code>\classXimera</code>	<code>\newtheorem{lemma}{\GetTranslation{Lemma}}</code>
	228 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{lemma}</code>
<code>formula (env.)</code>	Formula	
	229 <code>\classXimera</code>	<code>\newtheorem{formula}{\GetTranslation{Formula}}</code>
	230 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{formula}</code>
<code>idea (env.)</code>	Idea	
	231 <code>\classXimera</code>	<code>\newtheorem{idea}{\GetTranslation{Idea}}</code>
	232 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{idea}</code>
<code>notation (env.)</code>	Notation	
	233 <code>\classXimera</code>	<code>\newtheorem{notation}{\GetTranslation{Notation}}</code>
	234 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{notation}</code>
<code>model (env.)</code>	Model	
	235 <code>\classXimera</code>	<code>\newtheorem{model}{\GetTranslation{Model}}</code>
	236 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{model}</code>
<code>observation (env.)</code>	Observation	
	237 <code>\classXimera</code>	<code>\newtheorem{observation}{\GetTranslation{Observation}}</code>
	238 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{observation}</code>
<code>proposition (env.)</code>	Proposition	
	239 <code>\classXimera</code>	<code>\newtheorem{proposition}{\GetTranslation{Proposition}}</code>
	240 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{proposition}</code>
<code>paradox (env.)</code>	Paradox	
	241 <code>\classXimera</code>	<code>\newtheorem{paradox}{\GetTranslation{Paradox}}</code>
	242 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{paradox}</code>
<code>procedure (env.)</code>	Procedure	
	243 <code>\classXimera</code>	<code>\newtheorem{procedure}{\GetTranslation{Procedure}}</code>
	244 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{procedure}</code>
<code>remark (env.)</code>	Remark	
	245 <code>\classXimera</code>	<code>\newtheorem{remark}{\GetTranslation{Remark}}</code>
	246 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{remark}</code>
<code>summary (env.)</code>	Summary	
	247 <code>\classXimera</code>	<code>\newtheorem{summary}{\GetTranslation{Summary}}</code>
	248 <code>\htXimera</code>	<code>\ConfigureTheoremEnv{summary}</code>

```

template (env.)    Template
249 <classXimera>    \newtheorem{template}{\GetTranslation{Template}}
250 <htXimera>       \ConfigureTheoremEnv{template}

warning (env.)    Warning
251 <classXimera>    \newtheorem{warning}{\GetTranslation{Warning}}
252 <htXimera>       \ConfigureTheoremEnv{warning}

```

2.4.4 Enumerate fixes

Make enumerate use a letter

```

253 <*classXimera>
254 \renewcommand{\theenumi}{\textup{(\alph{enumi})}}
255 \renewcommand{\labelenumi}{\theenumi}
256 \renewcommand{\theenumii}{\textup{(\roman{enumii})}}
257 \renewcommand{\labelenumii}{\theenumii}
258 </classXimera>

259 <*cfgXimera>
260 \catcode'\:=11
261 % Insert <section> around thebibliography
262 \ConfigureEnv{thebibliography}{\ifvmode\IgnorePar\fi \EndP \HCode{<section role="doc-bibliography">}}
263 % now configure thebibliography to produce a description list
264 % \en:bib insertes delimiters for particular bibitems. at the beginning, it is empty, as then
265 % it is then defined to insert the delimiter after the first bibitem
266 \ConfigureList{thebibliography}%
267   {\ifvmode\IgnorePar\fi\EndP\HCode{<dl><dt>}\let\en:bib=\empty}% opening tags
268   {\ifvmode\IgnorePar\fi\EndP\HCode{</dd></dl>}} % closing tags
269   {\en:bib\def\en:bib{\ifvmode\IgnorePar\fi\HCode{</dd><dt>}}}% at the bibitem
270   {\HCode{</dt><dd>}}}% after biblabel
271 \catcode'\:=12
272 \Css{.thebibliography dl {
273     display: grid;
274     grid-auto-columns: min-content 1fr;
275     grid-auto-flow: column;
276 }}
277 \Css{.thebibliography dt {
278     grid-column: 1;
279     margin-bottom: 0.5em;
280 }}
281 \catcode'\:=11
282 \ConfigureList{enumerate}%
283   {\EndP\HCode{<ol \a:enumerate:\space
284     class="enumerate\expandafter\the\csname @enumdepth\endcsname"
285     \a:LRdir
286     >}\PushMacro\end:itm
287 \global\let\end:itm=\empty
288 }
289   {\PopMacro\end:itm \global\let\end:itm \end:itm
290 %
291 \EndP\HCode{</li></ol>}}\ShowPar
292 }
293   {\end:itm \gdef\end:itm{\EndP\Tg</li>}\DeleteMark
294 }
295   {{\Configure{Link}{li}{\a:enumerate id=}}}%
296   \let\EndLink=\empty\par\ShowPar
297 \AnchorLabel }%
298 }
299 \catcode'\:=12
300 </cfgXimera>

```

2.4.5 Proofs

proof (env.) A mathematical proof environment.


```

301 \classXimera
302 \renewcommand{\qedsymbol}{\blacksquare}
303 \renewenvironment{proof}[1][\proofname]
304 {\begin{trivlist}\item[\hskip \labelsep \itshape \bfseries #1]{\hspace{2ex}}}
305 {\qed\end{trivlist}}
306 \endclassXimera
307 \htXimera
308 % Mmm, (why) do we want/need this ...?
309 \ConfigureTheoremEnv{proof}
310 \ConfigureEnv{proof}{\ifvmode\IgnorePar\fi\EndP\HCode{\div class="proof">}}
311 \ConfigureList{trivlist}{\ifvmode\IgnorePar\fi\EndP}{\}{\}{}
312 {\ifvmode\IgnorePar\fi\EndP\HCode{</div>}}{\}{\}{}
313 \endhtXimera

```

2.4.6 Problem environments

These are problem environment decorations (these should be user invoked, not default). The decoration for these environments were inspired by <http://tex.stackexchange.com/questions/11098/nice-formatting-for-theorems>

```

314 \classXimera
315 \newcommand{\hang}{% top theorem decoration
316 \begin{group}
317 \setlength{\unitlength}{.005\linewidth}% \linewidth/200
318 \begin{picture}(0,0)(1.5,0)%
319 \linethickness{1pt} \color{black!50}%
320 \put(-3,2){\line(1,0){206}}% Top line
321 \multido{\iA=2+-1,\iB=50+-10}{5}{% Top hangs
322 \color{black!\iB}%
323 \put(-3,\iA){\line(0,-1){1}}% Top left hang
324 \put(203,\iA){\line(0,-1){1}}% Top right hang
325 }%
326 \end{picture}%
327 \endgroup%
328 }%
329 \newcommand{\hung}{% bottom theorem decoration
330 \nobreak
331 \begin{group}
332 \setlength{\unitlength}{.005\linewidth}% \linewidth/200
333 \begin{picture}(0,0)(1.5,0)%
334 \linethickness{1pt} \color{black!50}%
335 \put(60,0){\line(1,0){143}}% Bottom line
336 \multido{\iA=0+1,\iB=50+-10}{5}{% Bottom hangs
337 \color{black!\iB}%
338 \put(-3,\iA){\line(0,1){1}}% Bottom left hang
339 \put(203,\iA){\line(0,1){1}}% Bottom right hang
340 \put(\iB,0){\line(60,0){10}}% Left fade out
341 }%
342 \end{picture}%
343 \endgroup%
344 }%

```

Configure environment configuration commands

The command `\problemNumber` contains all the format code to determine the number (and the format of the number) for any of the problem environments.

```

345 \MakeCounter{Iteration@probCnt}
346 \MakeCounter{problem}
347 \newcommand{\problemNumber}{
348 % First we determine if we have a counter for this question depth level.
349 \ifcsname c@depth\Roman{problem@Depth}Count\endcsname% Check to see if counter exists
350 %If so, do nothing.
351 \else
352 %If not, create it.
353 \expandafter\newcounter{depth\Roman{problem@Depth}Count}

```

```

354 \expandafter\setcounter{depth\Roman{problem@Depth}Count}{0}
355 \fi
356
357 \expandafter\stepcounter{depth\Roman{problem@Depth}Count}
358 \arabic{depthICount}% The first problem depth, what use to be |\theproblem|.
359
360 \forloop{Iteration@probCnt}{2}{\arabic{Iteration@probCnt} < \numexpr \value{problem@Depth} +
361 .\expandafter\arabic{depth\Roman{Iteration@probCnt}Count}% Get the problem number of the next
362 }
363 }
364 %%%% Configure various problem environment commands
365 \Make@Counter{problem@Depth}
366 %%%% Configure environments start content
367 \newcommand{\problemEnvironmentStart}[2]{%
368 \stepcounter{problem@Depth}% Started a problem, so we've sunk another problem layer.
369 \def\spaceatend{#1}%
370 \begin{trivlist}%
371 \item[\hskip\labelsep\sffamily\bfseries\GetTranslation{#2} \problemNumber% Determine the corr
372 ]%
373 \slshape
374 }
375 %%%% Configure environments end content %%%%
376 \newcommand{\problemEnvironmentEnd}{%This configures all the end content for a problem.
377 \stepcounter{problem@Depth}
378 \ifcsname c@depth\Roman{problem@Depth}Count\endcsname
379 \expandafter\ifnum\expandafter\value{depth\Roman{problem@Depth}Count}>0
380 \expandafter\setcounter{depth\Roman{problem@Depth}Count}{0}
381 \fi
382 \fi
383 \addtocounter{problem@Depth}{-2}% Exited a problem so we've exited a problem layer. Need -2 b
384 \ifhandout
385 \ifnewpage
386 \newpage
387 \fi
388 \fi
389 \end{trivlist}
390 }
391 %% Add a simple command that handles all the problem creation aspects:
392 \newcommand{\createProblemEnv}[2]{% This is a nice command to define a new problem-like envi
393 \newenvironment{#1}[1][2in]%
394 {%Env start code
395 \problemEnvironmentStart{#1}{#2}
396 }
397 {%Env end code
398 \problemEnvironmentEnd
399 }
400 }
401
402 %%%% Now populate the old environment names
403 %
404 % Old environments were "problem", "exercise", "exploration", and "question".
405 % Note that you can add content to the start/end code on top of these base code pieces if you
406 %
407 % These definitions will be overwritten in ximera.4ht !
408
409 \createProblemEnv{problem}{Problem}
410 \createProblemEnv{exercise}{Exercise}
411 \createProblemEnv{exploration}{Exploration}
412 \createProblemEnv{question}{Question}
413 \</classXimera>
414 \<htXimera>
415 \newcounter{identification}
416 \setcounter{identification}{0}

```

```

417 \newcommand{\ConfigureQuestionEnv}[2]{%
418 \renewenvironment{#1}{
419 }
420 {
421 }%
422 \ConfigureEnv{#1}
423 {
424 % \ifnumberedProblems% The code below is all to generate online problem numbering if opti
425 % \stepcounter{problem@Depth}% Started a problem, so we've sunk another problem layer.
426 % \ifcsname c@depth\Roman{problem@Depth}Count\endcsname
427 % \else
428 % \expandafter\newcounter{depth\Roman{problem@Depth}Count}
429 % \expandafter\setcounter{depth\Roman{problem@Depth}Count}{0}
430 % \fi
431 % \expandafter\stepcounter{depth\Roman{problem@Depth}Count}
432 % \def\problemNumDisp{
433 % \arabic{depthICount}% Top Level Problem Number: X.1.1.1.1 Number.
434 % \ifcsname c@depthIICount\endcsname\ifnum\value{problem@Depth}>1 .\arabic{depthIICount}\fi
435 % \ifcsname c@depthIIICount\endcsname\ifnum\value{problem@Depth}>2 .\arabic{depthIIICount}\fi
436 % \ifcsname c@depthIVCount\endcsname\ifnum\value{problem@Depth}>3 .\arabic{depthIVCount}\fi
437 % \ifcsname c@depthVCount\endcsname\ifnum\value{problem@Depth}>4 .\arabic{depthVCount}\fi
438 % \fi\fi\fi\fi
439 % }
440 % \else
441 % \def\problemNumDisp{}% Otherwise don't display a problem number.
442 % \fi
443 \stepcounter{identification}
444 \ifvmode
445 \IgnorePar
446 \fi
447 \EndP
448 \HCode{<div role="article" class="problem-environment #1" id="problem\arabic{identification}"}
449 }
450 {
451 \stepcounter{problem@Depth}
452 \ifcsname c@depth\Roman{problem@Depth}Count\endcsname
453 \expandafter\ifnum\expandafter\value{depth\Roman{problem@Depth}Count}>0
454 \expandafter\setcounter{depth\Roman{problem@Depth}Count}{0}
455 \fi
456 \fi
457 \addtocounter{problem@Depth}{-2}% Exited a problem so we've exited a problem layer. Need -2
458 \ifvmode
459 \IgnorePar
460 \fi
461 \EndP
462 \HCode{</div>}\IgnoreIndent
463 }{}{}%
464 }
465
466 \ConfigureQuestionEnv{problem}{Problem}
467 \ConfigureQuestionEnv{exercise}{Exercise}
468 \ConfigureQuestionEnv{question}{Question}
469 \ConfigureQuestionEnv{exploration}{Exploration}
470
471 \ifdefined\xmNotHintAsExpandable
472 \ConfigureQuestionEnv{hint}{hint} % 2024: hint is no longer a 'question-environment'.
473 \fi
474 </htXimera>

```

2.4.7 Hints

`hint (env.)` Hint environments can be embedded inside problems.

```
475 <*classXimera>
```

Create a counter that will track how deeply nested the current hint is

```
476 \newcounter{hintLevel}
477 \setcounter{hintLevel}{0}
```

Create an empty shell to renew

```
478 \newenvironment{hint}{}{}
```

Now we renew the environment as needed, this should allow support for any transition code that treats some parts as a "handout" and some parts as non-handout. renewing the environment on the fly is a bit hacky.

```
479 \renewenvironment{hint}
480 {
481   \ifhandout
482     \setbox0\vbox\bgroup
483   \else
484     \begin{trivlist}\item[\hskip \labelsep\small\slshape\bfseries \GetTranslation{Hint}:\hspace{1ex}]
485     \small\slshape
486   \fi
487   \stepcounter{hintLevel}
488 }
489 {
490   \ifhandout
491     \egroup\ignorespacesafterend
492   \else
493     \end{trivlist}
494   \fi
495   \addtocounter{hintLevel}{-1}
496 }
497
498 \ifhints
499 \renewenvironment{hint}{
500   \begin{trivlist}\item[\hskip \labelsep\small\slshape\bfseries \GetTranslation{Hint}:\hspace{1ex}]
501   \small\slshape
502 }
503 {
504   \end{trivlist}
505 }
506 \fi
507
508 </classXimera>
```

2.4.8 Solution

`solution (env.)` The solution to a problem.

```
509 <*classXimera>
510 %% solution environment
511 \ifhandout % what follows is handout behavior
512 \newenvironment{solution}%
513   {%
514     \setbox0\vbox\bgroup
515   }
516   {%
517     \egroup
518   }
519 \else
520 \newenvironment{solution}%
521   {%
522     \begin{trivlist}
523     \item[\hskip \labelsep\bfseries \GetTranslation{Solution}:\hspace{2ex}]
524     }
525     % %% line at the bottom}
526     {
527     \end{trivlist}
528     % (202410: no longer \par\addvspace{.5ex}\nobreak\noindent\hung
```

```

529      }
530 \fi
531
532
533
534 \end{classXimera}

```

2.4.9 Code listing environments

code (*env.*) A code answer environment You cannot use Environ with the fancyvrb/listings package if you want nested environments.

```

535 \begin{classXimera}
536 \DefineVerbatimEnvironment{code}{Verbatim}{numbers=left,frame=lines,label=Code,labelposition=left}
537 \end{classXimera}

```

python (*env.*) A python answer environment You cannot use Environ with the fancyvrb/listings package if you want nested environments

```

538 \begin{classXimera}
539 \DefineVerbatimEnvironment{python}{Verbatim}{numbers=left,frame=lines,label=Python,labelposition=left}
540 \end{classXimera}

```

javascriptCode (*env.*) A JavaScript answer environment Unfortunately the name javascript is already used for the actual, executed (!) JavaScript interactive. environments

```

541 \begin{classXimera}
542 \DefineVerbatimEnvironment{javascriptCode}{Verbatim}{numbers=left,frame=lines,label=JavaScript,labelposition=left}
543 \end{classXimera}
544 \begin{classXimera}
545 \renewenvironment{javascriptCode}{\NoFonts}{\EndNoFonts}
546 \ScriptEnv{javascriptCode}{\stepcounter{identification}\ifvmode \IgnorePar\fi \EndP\HCode{<div>
547 \end{classXimera}

```

On the web, translate verbatim and lstlisting blocks into <pre> elements.

```

548 %\begin{classXimera}
549 %\ConfigureEnv{verbatim}{\ifvmode\IgnorePar\fi\EndP\HCode{<pre style="white-space: pre; background-color: #f0f0f0; padding: 10px; border: 1px solid #ccc;">
550 %\ConfigureEnv{lstlisting}{\ifvmode\IgnorePar\fi\EndP\HCode{<pre>}}{\ifvmode\IgnorePar\fi\EndP}
551 %\end{classXimera}
552 %

```

2.4.10 Dialogues

dialogue (*env.*) A dialogue between people.

```

553 \begin{classXimera}
554 \newenvironment{dialogue}{%
555   \renewcommand\descriptionlabel[1]{\hspace{\labelsep}\textbf{##1:}}
556   \begin{description}%
557 }{%
558   \end{description}%
559 }
560 \end{classXimera}

```

On the web, the resulting <dl> should have an appropriate class set.

```

561 \begin{classXimera}
562 \renewenvironment{dialogue}{\begin{description}}{\end{description}}
563
564 \ConfigureList{dialogue}{%
565   {\EndP\HCode{<dl \a:LRdir class="dialogue">}}%
566   \PushMacro\end:itm
567 \global\let\end:itm=\empty
568   {\PopMacro\end:itm \global\let\end:itm \end:itm \end:itm
569 \EndP\HCode{</dd></dl>}}\ShowPar}
570   {\end:itm \global\def\end:itm{\EndP\Tg</dd>}\HCode{<dt
571     class="actor">}\bgroup \bf}
572   {\egroup\EndP\HCode{</dt><dd\Hnewline class="speech">}}
573 \end{classXimera}

```

2.4.11 Instructor notes

```

574 \classXimera)
575
576 %% instructor intro/instructor notes
577 %%
578 \ifhandout % what follows is handout behavior
579 \ifinstructornotes
580 \newenvironment{instructorIntro}%
581     {%
582     \begin{trivlist}
583     \item[\hskip \labelsep\bfseries \GetTranslation{Instructor Introduction}:\hspace{2ex}]
584     }
585     % %% line at the bottom}
586     {
587     \end{trivlist}
588     \par\addvspace{.5ex}\nobreak\noindent\hung
589     }
590 \else
591 \newenvironment{instructorIntro}%
592     {%
593     \setbox0\vbox\bgroup
594     }
595     {%If this mysteriously starts breaking
596     % remove \ignorespacesafterend
597     \egroup\ignorespacesafterend
598     }
599     \fi
600 \else% for handout, so what follows is default
601 \ifinstructornotes
602 \newenvironment{instructorIntro}%
603     {%
604     \setbox0\vbox\bgroup
605     }
606     {%
607     \egroup
608     }
609     \else
610     \newenvironment{instructorIntro}%
611     {%
612     \begin{trivlist}
613     \item[\hskip \labelsep\bfseries \GetTranslation{Instructor Introduction}:\hspace{2ex}]
614     }
615     % %% line at the bottom}
616     {
617     \end{trivlist}
618     \par\addvspace{.5ex}\nobreak\noindent\hung
619     }
620     \fi
621 \fi
622
623
624
625
626 %% instructorNotes environment
627 \ifhandout % what follows is handout behavior
628 \ifinstructornotes
629 \newenvironment{instructorNotes}%
630     {%
631     \begin{trivlist}
632     \item[\hskip \labelsep\bfseries \GetTranslation{Instructor Notes}:\hspace{2ex}]
633     }
634     % %% line at the bottom}
635     {

```

```

636 \end{trivlist}
637 \par\addvspace{.5ex}\nobreak\noindent\hung
638 }
639 \else
640 \newenvironment{instructorNotes}%
641 {%
642 \setbox0\vbox\bgroup
643 }
644 {%
645 \egroup
646 }
647 \fi
648 \else% for handout, so what follows is default
649 \ifinstructornotes
650 \newenvironment{instructorNotes}%
651 {%
652 \setbox0\vbox\bgroup
653 }
654 {%
655 \egroup
656 }
657 \else
658 \newenvironment{instructorNotes}%
659 {%
660 \begin{trivlist}
661 \item[\hskip \labelsep\bfseries \GetTranslation{Instructor Notes}:\hspace{2ex}]
662 }
663 % %% line at the bottom}
664 {
665 \end{trivlist}
666 \par\addvspace{.5ex}\nobreak\noindent\hung
667 }
668 \fi
669 \fi
670
671 </classXimera>

```

2.4.12 Foldable

The package `mdframed` is used to make pretty foldable, but the `amsthm/mdframed` conflict also messes up the `.jax` file so we don't load `mdframed` when performing the `xake` step. But even the below isn't enough to fix this.

```

672 %\iftikzexport\else\RequirePackage[framemethod=TikZ]{mdframed}\fi
foldable (env.) Does it fold?
673 <{*classXimera>
674
675 \colorlet{textColor}{black} % since textColor is referenced below
676 \colorlet{background}{white} % since background is referenced below
677
678 % The core environments. Find results in 4ht file.
679 %% pretty-foldable
680 %\iftikzexport
681 \newenvironment{foldable}{%
682 }{%
683 }
684 %\else
685 %\renewmdenv[
686 % font=\upshape,
687 % outerlinewidth=3,
688 % topline=false,
689 % bottomline=false,
690 % leftline=true,
691 % rightline=false,

```

```

692 % leftmargin=0,
693 % innertopmargin=0pt,
694 % innerbottommargin=0pt,
695 % skipbelow=\baselineskip,
696 % linecolor=textColor!20!white,
697 % fontcolor=textColor,
698 % backgroundcolor=background
699 %]{foldable}%
700 %\fi
701
702 %% pretty-expandable
703 %\iftikzexport
704 %% Overwritten in .4ht, but probably also in accordion!
705 \ifdefined\xmNotExpandableAsAccordion
706 \newenvironment{expandable}{}{}
707 \else
708 \newenvironment{expandable}[2]{}{}
709 \fi
710 \else
711 %\newmdenv[
712 % font=\upshape,
713 % outerlinewidth=3,
714 % topline=false,
715 % bottomline=false,
716 % leftline=true,
717 % rightline=false,
718 % leftmargin=0,
719 % innertopmargin=0pt,
720 % innerbottommargin=0pt,
721 % skipbelow=\baselineskip,
722 % linecolor=black,
723 %]{expandable}%
724 %\fi
725
726 \newcommand{\unfoldable}[1]{#1}
727
728 \</classXimera>

```

On the web, foldable is a native HTML5 disclosure widget: `<details open>` (foldables start expanded, matching the historical chevron-button behavior) whose `<summary>` is the leading `\unfoldable` content, so the teaser stays visible when the reader folds the rest away. Native details/summary gives keyboard operability and expanded/collapsed screen-reader announcements with no JavaScript at all. The class is "foldable-details" rather than "foldable" on purpose: the legacy frontend attaches its jQuery folding logic (chevron button, `children().hide()`) to every ".foldable" element, which would fight the native widget and hide the summary.

```

729 <*htXimera>
730 \newif\ifxm@in@foldable
731 \newif\ifxm@foldable@fresh
732 \renewenvironment{foldable}{%
733   \stepcounter{identification}%
734   \ifvmode \IgnorePar\fi \EndP%
735   \HCode{<details open="open" id="foldable\arabic{identification}" class="foldable-details">}
736   \xm@in@foldabletrue
737   \global\xm@foldable@freshtrue
738 }{%
739   \global\xm@foldable@freshfalse
740   \ifvmode\IgnorePar\fi\EndP\HCode{</details>}\IgnoreIndent
741 }
742
743 \ifdefined\xmNotExpandableAsAccordion
744 \renewenvironment{expandable}{\stepcounter{identification}\ifvmode \IgnorePar\fi \EndP\HCode-
745 \fi

```



```

746
747 % The first \unfoldable at the start of a foldable becomes the <summary>.
748 % HTML allows only one summary per details, so any later \unfoldable in the
749 % same foldable cannot stay visible when folded; it degrades to a plain
750 % inline span and warns the author. Outside a foldable, \unfoldable remains
751 % a plain span as before.
752 \renewcommand{\unfoldable}[1]{%
753   \ifxm@in@foldable
754     \ifxm@foldable@fresh
755       \global\xm@foldable@freshfalse
756       \ifvmode \IgnorePar\fi \EndP%
757       \HCode{<summary class="unfoldable">}#1\HCode{</summary>\Hnewline}%
758     \else
759       \PackageWarning{ximera}{Only the first \string\unfoldable\space at the start of a foldable}
760       \HCode{<span class="unfoldable">}#1\HCode{</span>}%
761     \fi
762   \else
763     \HCode{<span class="unfoldable">}#1\HCode{</span>}%
764   \fi}
765 \end{htXimera}

```

2.4.13 Leashes

`\leash (env.)` Put content inside a scrollable box.

```

766 \classXimera
767
768 \newenvironment{leash}[1]{%
769 }{%
770 }
771
772
773 \end{classXimera}
774 \end{htXimera}
775 \renewenvironment{leash}[1]{\ifvmode \IgnorePar\fi \EndP\HCode{<div style="overflow: auto; height: 100px; border: 1px solid black; padding: 5px; margin: 5px 0;">}#1\HCode{</div>}}{\ifvmode \IgnorePar\fi \EndP}
776 \end{htXimera}

```

2.5 Document metadata

2.5.1 Metadata

To encourage authors to include relevant parseable metadata in the preamble, we define some currently ignored commands.

`\license` In the preamble, use `\license` with an SPDX license expression.

```

777 \classXimera
778 \newcommand{\license}{\excludecomment}
779 \end{classXimera}

```

`\acknowledgement` In the preamble, use `\acknowledgement` to credit others who contributed to the intellectual content beside the author.

```

780 \classXimera
781 \newcommand{\acknowledgement}{\excludecomment}
782 \end{classXimera}

```

`\tag` In the preamble, a `\tag` provides a free-form taxonomy.

```

783 \classXimera
784 \renewcommand{\tag}{\excludecomment}
785 \end{classXimera}

```

On the HTML side, we mark the file as the appropriate kind of object—either activity or xourse.

```

786 \end{htXourse}
787 % Mark this as a xourse file
788 \Configure{@HEAD}{\HCode{<meta name="description" content="xourse" />\Hnewline}}
789 \end{htXourse}

```

2.5.2 Abstract

`abstract (env.)` Every activity should include a short abstract.

```
790 \let\abstract\relax
791 \let\endabstract\relax
792 % Use of environ package, may want to find a better way.
793 % see the messing around with \theabstract in title.dtx ... Is this really needed/wanted?
794 \NewEnviron{abstract}{\protected@xdef\theabstract{\BODY}}
795 \let\abstract\relax
796 \let\endabstract\relax
```

The abstract has been stored in `\theabstract` and should be emitted as a div. The code below is required for the abstract to show online.

```
797 \let\abstract\relax
798 \ifvmode\IgnorePar\fi\EndP
799 \ConfigureEnv{abstract}{\ifvmode\IgnorePar\fi\EndP\HCode{\Hnewline<div class="abstract">}\par}
800 \let\abstract\relax
801 \let\endabstract\relax
802 \RenewEnviron{abstract}{\BODY}
803 \let\abstract\relax
804 \let\endabstract\relax
```

2.5.3 Titles and authors

2.5.4 Authors

`\author` Activities have authors. Warn the user if no author is provided.

```
804 \let\author\relax
805 \let\emptyauthor\@author
806 \def\@authorfootnote{\gdef\@thefnmark{}\@footnotetext}
807 \def\author#1{\gdef\@author{#1}}
808 \def\@author{\@latex@warning@no@line{No \noexpand\author given}}
809 \let\author\relax
```

Include author name in meta tags

```
810 \let\author\relax
811 \Configure{@HEAD}{\HCode{<meta name="author" content="}\@author\HCode{ />}\Hnewline}}
812 \let\author\relax
```

The `\and` command would emit tabular environments which really should not appear in a meta tag.

```
813 \let\and\relax
```

2.5.5 Title

`\title` Activities have titles.

```
814 \let\title\relax
815 \let\endtitle\relax
816 \newcommand{\title}[1]{}{\protected@xdef\prettitle{#1}\protected@xdef\@title{#1}}
817 \let\title\relax
818 \let\endtitle\relax
819 \let\title\relax
820 \newcounter{titlenumber}
821 \renewcommand{\thetitle}{\arabic{titlenumber}}
822 \renewcommand{\thesection}{\arabic{titlenumber}} %% Makes section numbers work
823 \setcounter{titlenumber}{0}
824 \let\title\relax
825 \newpagestyle{main}{
826 \sethead[\textsl{\ifnumbers\thetitle\hspace{1em}\fi\@title}]{}{} % even
827 {}{}{\textsl{\ifnumbers\thetitle\hspace{1em}\fi\@title}} % odd
828 \setfoot[\thepage]{}{} % even
829 {}{}{\thepage} % odd
830 }
831 \pagestyle{main}
```

`\maketitle` In a ximera document, redefine `\maketitle` and put them in a table of contents. The `\phantomsection` is to fix the hrefs.

```

832 \renewcommand\maketitle{%
833   \addtocounter{titlenumber}{1}%
834   {\flushleft\large\bfseries \@pretitle\par\vspace{-1em}}
835   {\flushleft\LARGE\bfseries {\ifnumbers\thetitlenumber\fi}{\ifnumbers\hspace{1em}\else\hspa
836   \phantomsection%
837   \ifnumbers\addcontentsline{toc}{section}{\thetitlenumber~\@title}\else\addcontentsline{toc
838   \vskip .6em\noindent\textit{theabstract}\setcounter{problem}{0}\setcounter{section}{0}\setco
839   %\ifnooutcomes\else\let\thefootnote\relax\footnote{Learning outcomes: \theoutcomes}\fi% Dep
840   \ifnoauthor\else\@authorfootnote{Author(s):~\@author}\fi
841   \aftergroup\@afterindentfalse
842   \aftergroup\@afterheading}
843
844 \ifnumbers
845 \setcounter{secnumdepth}{2}
846 \renewcommand{\thesection}{\arabic{titlenumber}.\arabic{section}}
847 \renewcommand{\thesubsection}{\arabic{titlenumber}.\arabic{section}.\arabic{subsection}}
848 \else
849 \setcounter{secnumdepth}{-2}
850 \fi
851
852 \def\activitystyle{}
853 \newcounter{sectiontitlenumber}
854 \setcounter{secnumdepth}{2}
855 \setcounter{tocdepth}{2}
856 \newcommand\chapterstyle{%
857   \def\activitystyle{activity-chapter}
858   \def\maketitle{%
859     \addtocounter{titlenumber}{1}%
860     {\flushleft\small\sffamily\bfseries\@pretitle\par\vspace{-1.5em}}%
861     {\flushleft\LARGE\sffamily\bfseries\thetitlenumber\hspace{1em}\@title \par
862     {\vskip .6em\noindent\textit{theabstract}\setcounter{problem}{0}\setcount
863     \par\vspace{2em}
864     \phantomsection\addcontentsline{toc}{section}{\textbf{\thetitlenumber\hspa
865   }}
866
867
868 \newcommand\sectionstyle{%
869   \def\activitystyle{activity-section}
870   \def\maketitle{%
871     \addtocounter{section}{1}
872     \setcounter{sectiontitlenumber}{\value{section}}
873     {\flushleft\small\sffamily\bfseries\@pretitle\par\vspace{-1.5em}}%
874     {\flushleft\Large\sffamily\bfseries\thetitlenumber.\thesectiontitlenumber\hspace{1em}\@t
875     {\vskip .6em\noindent\textit{theabstract}\setcounter{subsection}{0}}%
876     \par\vspace{2em}
877     \phantomsection\addcontentsline{toc}{section}{\thetitlenumber.\thesectiontitlenumber\hspa
878   \renewcommand\section{\@startsection{subsection}{2}{\z@}%
879     {-3.25ex\@plus -1ex \@minus -.2ex}%
880     {1.5ex \@plus .2ex}%
881     {\normalfont\large\bfseries}}
882
883   \renewcommand\subsection{\@startsection{subsubsection}{3}{\z@}%
884     {-3.25ex\@plus -1ex \@minus -.2ex}%
885     {1.5ex \@plus .2ex}%
886     {\normalfont\normalsize\bfseries}}
887
888 }}
889
890
891 \iftikzexport%% allows xake to handle \chapterstyle and \sectionstyle
892 \renewcommand\chapterstyle{\def\activitystyle{chapter}}

```

```

893 \renewcommand\sectionstyle{\def\activitystyle{section}}
894 \else
895 \fi
896
897 \endclassXimera

```

Eliminate some formatting that we'll handle later with CSS

```

898 \beginhtXimera
899 \renewcommand{\maketitle}{}
900 \endhtXimera

```

2.5.6 Only in HTML or PDF

Ximera provides several techniques to display some content only in the PDF, or only online. The `prompt` environment can be used to hide the data-entry part of a problem from the PDF: it's contents only get displayed online.

The lower level commands `\pdfOnly` and `\htmlOnly` also limit the output to either PDF or online, similarly to the environments `onlyPdf` and `onlyHtml`.

If `\xmPrintHtmlOnlyAlsoInPdf` is set, the online/html only things are printed in the PDF anyway (e.g. for review).

Unfortunately it is not possible in $\text{\LaTeX}$ to have a command and an environment with the same name. We opted for the above (confusing...) names.

For backward compatibility, the deprecated environment `onlineOnly` is identical to `onlyHtml`.

For more advanced usage also commands `\ifonline` and `ifonlineTF` are provided.

The technique used to distinguish between the PDF-version and the online HTML-version is always the existence of the TeX4ht macro `\HCode`. Older distinctions such as `\ifxake`, `ifhandout` or `\iftikzexport` should no longer be used for this purpose.

`prompt (env.)` The prompt part for mathmode

```

901 \beginclassXimera
902 \ifxake
903     \newenvironment{prompt}{}{}
904 \else
905 \ifhandout
906     \NewEnviron{prompt}{}
907     % Breaks when put in mathmode ?
908     % \newenvironment{prompt}{\suppress}{\endsuppress}
909 \else
910     \newenvironment{prompt}{\bgroup\color{gray!50!black}}{\egroup}
911 \fi
912 \fi

```

`onlyHtml (env.)` Only display online

`onlyPdf (env.)` Only display in the PDF

`onlineOnly (env.)` Only display online (deprecated: use `onlyHtml` instead)

```

913 \ifdefined\HCode
914     \newenvironment{onlyPdf}{\setbox0\vbox\bgroup}{\egroup}
915     \newenvironment{onlyHtml}{\bgroup}{\egroup}
916     \newenvironment{onlineOnly}{\bgroup}{\egroup}
917 \else
918     \newenvironment{onlyPdf}{\bgroup}{\egroup}
919     \ifdefined\xmPrintHtmlOnlyAlsoInPdf
920         \newenvironment{onlyHtml}{\bgroup\color{red!50!black}}{\egroup}
921         \newenvironment{onlineOnly}{\bgroup\color{red!50!black}}{\egroup}
922     \else
923         \newenvironment{onlyHtml}{\setbox0\vbox\bgroup}{\egroup}
924         \newenvironment{onlineOnly}{\setbox0\vbox\bgroup}{\egroup}
925     \fi
926 \fi
927

```

`\htmlOnly` Only display online

`\pdfOnly` Only display in the PDF

```

928
929 \ifdefined\HCode
930 \newcommand{\pdfOnly}[1]{ }
931 \newcommand{\htmlOnly}[1]{#1}
932 \else
933 \ifdefined\xmPrintHtmlOnlyAlsoInPdf
934 \newcommand{\pdfOnly}[1]{#1}
935 \newcommand{\htmlOnly}[1]{\bgroup\color{red!50!black}#1\egroup}
936 \else
937 \newcommand{\pdfOnly}[1]{#1}
938 \newcommand{\htmlOnly}[1]{ }
939 \fi
940 \fi
941
\ifonline Only execute online (ie in HTML version)
\ifonlineTF Different output online vs PDF
942 % An alternative for \pdfOnly/\begin{htmlOnly} :
943 % Usage: Hello \ifonlineTF{online reader}{PDF reader}
944 \providecommand{\ifonlineTF}[2]{\htmlOnly{#1}\pdfOnly{#2}}
945 \newif{\ifonline}
946 \ifdefined\HCode
947 \onlinetrue
948 \else
949 \onlinefalse
950 \fi
951 \end{classXimera}

```

2.5.7 Learning Outcomes

```

952 \classXimera
953 \newcommand{\preOutcomeLine}{\item }
954 \newcommand{\postOutcomeLine}{ }
955 \newcommand{\preOutcomeBlock}{After completing this content, students should be able to: \begin{itemize}
956 \newcommand{\postOutcomeBlock}{\end{itemize} So go forth and learn!}
957
958 \newcommand{\outcomeHeader}{Goals for this Section}
959 \htmlOnly{
960 \newcommand{\outcomeBlock}{\ifvmode\IgnorePar\fi\EndP\HCode{<div class="outcomeHead">} \outcomeHeader}
961 }
962
963
964 \newwrite\outcomefile
965 \immediate\openout\outcomefile=\jobname.oc
966 \newcommand{\outcome}[1]{%
967 \immediate\write\outcomefile{\expandafter\unexpanded\expandafter{\preOutcomeLine #1} \outcomeBlock}
968 }
969
970 \newcommand{\displayOutcomes}[1][ ]{%
971 \immediate\closeout\outcomefile
972 \IfFileExists{\currfiledir\currfilebase.oc}{
973 \htmlOnly{\outcomeBlock}
974 \expandafter\preOutcomeBlock
975 \input{\currfiledir\currfilebase.oc}
976 \postOutcomeBlock
977 \htmlOnly{\ifvmode\IgnorePar\fi\EndP\HCode{</div>}}
978 }
979 {
980 \IfFileExists{\currfilebase.oc}{
981 \htmlOnly{\outcomeBlock}
982 \expandafter\preOutcomeBlock
983 \input{\currfilebase.oc}
984 \postOutcomeBlock
985 \htmlOnly{\ifvmode\IgnorePar\fi\EndP\HCode{</div>}}

```

```

986     }
987     {
988         No outcome file found.
989     }
990 }
991 }
992 %
993 </classXimera>

```

These can appear in either the preamble or in problem environments. with pdflatex, we produce the .oc file which includes ALL the outcomes; in the tex4ht world, we just produce spans for the specific outcomes.

```

994 <*cfgXimera>
995 \renewcommand{\outcome}[1]{
996     \Configure{@HEAD}{\HCode{<meta name="learning-outcome" content="#1"/>\Hnewline}}
997 }
998 % Sometimes there are no outcomes at all
999 \IfFileExists{\jobname.oc}{\input{\jobname.oc}}{}
1000
1001 \renewcommand{\outcome}[1]{%
1002     \HCode{<span class="learning-outcome">#1</span>}
1003 }
1004 </cfgXimera>

```

2.5.8 Labels and references

\label Labels and refs both generate anchors. A **\label** can be referenced from any file in the xourse.

```

1005 <*htXimera>
1006 \let\oldlabel\label
1007 \renewcommand{\label}[1]{\oldlabel{#1}\HCode{<a class="ximera-label" id="#1"></a>}}
1008 </htXimera>

```

\ref A **\ref** can connect one T_EX file to another if they are in the same xourse.

```

1009 <*htXimera>
1010 \renewcommand{\ref}[1]{\HCode{<a class="reference" href="#1">#1</a>}}
1011 </htXimera>

```

2.6 Images

2.6.1 Images

image (env.) Place images inside an **image** environment. On paper, this centers the image. On the web, this provides additional benefits. Base graphicspath, default '/xmPictures'. Can only be changed BEFORE loading ximera.cls!

```

1012 <*classXimera>
1013 % Provide a default graphicspath
1014 % (somewhat tricky: an activity can be included in a xourse in a wildly different path !)
1015 % Suggested convention: put all images in i /pictures folder in the root of your project
1016 \providecommand{\xmDefaultGraphicsPath}{/xmPictures}
1017 \graphicspath{ %% When looking for images,
1018 {./} %% look here first,
1019 {.\xmDefaultGraphicsPath/} %% then look for a pictures folder,
1020 {..\xmDefaultGraphicsPath/} %% then look for a pictures folder,
1021 {../../xmDefaultGraphicsPath/} %% then look for a pictures folder,
1022 {../../../../xmDefaultGraphicsPath/} %% then look for a pictures folder,
1023 }
1024 %\newenvironment{image}[1][\begin{center}]{\end{center}}
1025 \NewEnviron{image}[1][3in]{%
1026     \begin{center}\resizebox{#1}{!}{\BODY}\end{center}% resize and center
1027 }
1028 </classXimera>

```

\alt Inside an **image** environment, **\alt** provides alt-text for assistive technology like screen-readers.

```

1029 <*classXimera>
1030 \newcommand{\alt}[1]{%
1031 </classXimera>

```

The `image` environment doesn't actually work in tex4ht as defined with `NewEnviron`; so this `renewenvironment` is needed. `image`-environment also gets formatted in a well, and when the user clicks on the image, it zooms in.

```

1032 <*htXimera>
1033 \newcounter{imagealt}
1034 \setcounter{imagealt}{0}
1035 \renewenvironment{image}[1][\stepcounter{imagealt}%
1036 \ifvmode \IgnorePar\fi \EndP%
1037 \HCode{<div class="image-environment" role="img" aria-labelledby="image-alt-\arabic{imagealt}>}%
1038 }{\HCode{</div>}}
1039 \renewcommand{\alt}[1]{\HCode{<div style="display: none;" id="image-alt-\arabic{imagealt}>>}}%
1040 </htXimera>
1041 <*cfgXimera>
1042 %% Although we accept many formats, SVG is preferred on the web.
1043 %% Since we have a different mechanism for producing |alt| text, we
1044 %% want to ignore tex4ht's own method fo producing alt text.
1045 %% 2024: is now in TeX4ht ...
1046 % \DeclareGraphicsExtensions{.jpg,.png,.gif,.svg}
1047 % \Configure{graphics*}
1048 % {svg}{
1049 % {\Configure{Needs}{File: \Gin@base.svg}\Needs{}}
1050 % \Picture[]{\csname Gin@base\endcsname.svg \csname a:Gin-dim\endcsname}%
1051 % }
1052 </cfgXimera>

```

This is a hack to kill `includegraphics` commands in `\documentclass{standalone}` files

```

1053 <*cfgXimera>
1054 \ifcsname ifstandalone\endcsname
1055 \ifstandalone
1056 \renewcommand\includegraphics[2][\fi
1057 \fi
1058 </cfgXimera>

```

PGF sometimes causes trouble, but we simply don't care in tex4ht mode.

```

1059 <*htXimera>
1060 \providecommand{\pgfsyspdfmark}[3]{%
1061 </htXimera>

```

2.6.2 TikZ export

2024: We DON NOT ANYMORE generate SVGs and PNGs for any TikZ images, via the “externalize” feature of TikZ.

Previously TikZ didn't compile natively into the website because of how the xake bake compilation works. In order to make Tikz work, you need to get the tool `mutool` on the machine that is performing `xake bake`.

```

1062 <*classXimera>
1063 % everything skipped, assume TeX4ht does the jbb now
1064 \ifdefined\reallyneverever
1065
1066 \ifdefined\HCode
1067 \tikzexporttrue
1068 \fi
1069
1070 \iftikzexport
1071 \usetikzlibrary{external}
1072
1073 \ifdefined\HCode
1074 % in htlatex, just include the svg files
1075 \def\pgfsys@imagesuffixlist{.svg}

```

```

1076
1077 \tikzexternalize[prefix=./,mode=graphics if exists]
1078 \else
1079 % in pdflatex, actually generate the svg files
1080 \tikzset{
1081 /tikz/external/system call={
1082 pdflatex \tikzexternalcheckshellescape
1083 -halt-on-error -interaction=batchmode
1084 -jobname "image" "\PassOptionsToClass{tikzexport}{ximera}\texsource";
1085 mutool draw -F svg \image.pdf > \image.svg ; % mutool adds "1" to filename ???
1086 mutool draw -o \image.svg \image.pdf ;
1087 mutool draw -r 150 -c rgbalpha -o \image.png \image.pdf ;
1088 ebb -x \image.png
1089 }
1090 }
1091 \tikzexternalize[optimize=false,prefix=./]
1092 \fi
1093
1094 \fi
1095 \fi
1096 \end{classXimera}

```

2.6.3 XKCD

`\xkcd` Reference an XKCD cartoon.

```

1097 \begin{classXimera}
1098 \newcommand{\xkcd}[1]{#1}
1099 \end{classXimera}

```

On the web, this should be an image linked to the actual XKCD website.

```

1100 \begin{htXimera}
1101 \renewcommand{\xkcd}[1]{\ifvmode \IgnorePar\fi \EndP\HCode{<img src="https://imgs.xkcd.com/c
1102 \end{htXimera}

```

2.7 Links

We put `hyperref` after all other packages because that is better.

```

1103 \begin{classXimera}
1104 % Don't use hyperref when using Tex4ht
1105 \ifdefined\HCode
1106 \RequirePackage{hyperref}
1107 \else
1108 \RequirePackage[pdfpagelabels,colorlinks=true,allcolors=blue!30!black]{hyperref}
1109 \pdfstringdefDisableCommands{\def\hskip{}}% quiets warning
1110 \fi
1111 \end{classXimera}

```

2.8 Interactives

2.8.1 Including widgets

`\includeinteractive` Cognate to `\includegraphics` but instead of a graphics file, accepts a `.js` file which will be loaded as an interactive widget.

```

1112 \begin{classXimera}
1113 \define@key{interactive}{id}{\def\interactiveid{#1}}
1114 \setkeys{interactive}{id=}
1115 \newcommand{\includeinteractive}[2][]{
1116 \setkeys*{interactive}{#1}%
1117 \ifthenelse{\equal{\interactiveid}{}}{\recordvariable{\interactiveid}}
1118 Interactive
1119 }
1120 \end{classXimera}

```



```

1121 <htXimera>
1122 \renewcommand{\includeinteractive}[2][\stepcounter{identification}\ifvmode \IgnorePar\fi \
1123 </htXimera>

```

2.8.2 Google Sheet

`\googleSheet` googleSheet command. Requires id, width, and height as arguments. optional arguments are gid for sheet ID and range for cell range. command definition

```

1124 <classXimera>
1125 % Google Spreadsheet link (read only)
1126 \newcommand{\googleSheet}[5]{%
1127   Google Spreadsheet link: \expandafter\url{\detokenize{https://docs.google.com/spreadsheets/
1128 }}
1129 </classXimera>
1130 <htXimera>
1131 \renewcommand{\googleSheet}[5]{%
1132   \ifthenelse{\equal{#4}{}}{%
1133     {\HCode{<iframe width="#2px" height="#3px" src="https://docs.google.com/spreadsheets/d/#
1134     {\ifthenelse{\equal{#5}{}}{%
1135       {\HCode{<iframe width="#2px" height="#3px" src="https://docs.google.com/spreadsheets/c
1136       {\HCode{<iframe width="#2px" height="#3px" src="https://docs.google.com/spreadsheets/c
1137     }}%
1138   }}%
1139 </htXimera>

```

2.8.3 Geogebra

`\geogebra` Geogebra command. Requires id, width, and height as arguments.

```

1140 <classXimera>
1141 %Geogebra link
1142 \newcommand{\geogebra}[3]{GeoGebra link: \url{https://www.geogebra.org/m/#1}}
1143 </classXimera>

```

Define keys for answer geogebra key=value pairs.

```

1144 <htXimera>
1145 \define@key{geogebra}{rc}[true]{\def\geo@rc{#1}}
1146 \define@key{geogebra}{sdz}[true]{\def\geo@sdz{#1}}
1147 \define@key{geogebra}{smb}[true]{\def\geo@smb{#1}}
1148 \define@key{geogebra}{stb}[true]{\def\geo@stb{#1}}
1149 \define@key{geogebra}{stbh}[true]{\def\geo@stbh{#1}}
1150 \define@key{geogebra}{ld}[true]{\def\geo@ld{#1}}
1151 \define@key{geogebra}{sri}[true]{\def\geo@sri{#1}}
1152 %set default key values
1153 \setkeys{geogebra}{rc=false,sdz=false,smb=false,stb=false,stbh=false,ld=false,sri=false}
1154 %command definition
1155 \renewcommand{\geogebra}[4][\%
1156   \setkeys{geogebra}{#1}% Set new keys
1157   \HCode{<iframe scrolling="no" src="https://www.geogebra.org/material/iframe/id/#2/width/#3
1158 </htXimera>

```

2.8.4 Desmos

`\desmos` Desmos command. Requires id, width, and height as arguments.

```

1159 <classXimera>
1160 \newcommand{\desmos}[3]{Desmos link: \url{https://www.desmos.com/calculator/#1}}
1161 \newcommand{\desmosThreeD}[3]{Desmos3D link: \url{https://www.desmos.com/3d/#1}}
1162 </classXimera>
1163 <htXimera>
1164 \catcode'\%=11
1165 \renewcommand{\desmos}[3]{\HCode{<iframe src="https://www.desmos.com/calculator/#1" width="1
1166 \catcode'\%=14
1167 \renewcommand{\desmosThreeD}[3]{\HCode{<iframe src="https://www.desmos.com/3d/#1" width="#2p
1168 </htXimera>

```

2.8.5 Graphs

`\graph` An embedded graph (in math mode).

```
1169 <*classXimera>
1170 \newcommand{\graph}[2][\text{Graph of } \mathbb{R}^2}
1171 </classXimera>

1172 <*htXimera>
1173 \renewcommand{\graph}[2][\HCode{<div class="graph" data-options="#1">}#2\HCode{</div>}}
1174 </htXimera>
```

2.8.6 Video

`\youtube` Youtube command. Requires id.

```
1175 <*classXimera>
1176 \newcommand{\youtube}[1]{YouTube link: \url{https://www.youtube.com/watch?v=#1}}
1177 </classXimera>

1178 <*htXimera>
1179 %% \renewcommand{\youtube}[1]{\ifvmode \IgnorePar\fi \EndP\HCode{<div class="video youtube-p
1180 % Fixes no-youtube-when-no-cookies-accepted. Class xmyoutube allows for css customization.
1181 \renewcommand{\youtube}[1]{\ifvmode \IgnorePar\fi \EndP\HCode{<iframe class="xmyoutube" src=
1182
1183 </htXimera>
```

Video commands are also emitted, slightly differently, when placed at top-level in a xourse file.

```
1184 <*htXourse>
1185 \renewcommand\youtube[1]{%
1186 \ifvmode \IgnorePar\fi \EndP\HCode{<a class="youtube" href="https://www.youtube.com/watch?v=
1187 }
1188 </htXourse>
```

2.8.7 JavaScript

`javascript (env.)` Code inside a javascript environment is printed on paper, but executed on the web.

```
1189 <*classXimera>
1190 \DefineVerbatimEnvironment{javascript}{Verbatim}{numbers=left,frame=lines,label=JavaScript,la
1191 </classXimera>

1192 <*htXimera>
1193 % for programming javascript
1194 \renewenvironment{javascript}{\NoFonts}{\EndNoFonts}
1195 \ScriptEnv{javascript}{\stepcounter{identification}\ifvmode \IgnorePar\fi \EndP\HCode{<div c
1196 </htXimera>
```

`\js` Code inside a `\js` macro is evaluated and replaced with its value.

```
1197 <*classXimera>
1198 \def\js#1{\mbox{\texttt{\detokenize{#1}}}}
1199 </classXimera>

1200 <*htXimera>
1201 \def\js#1{\stepcounter{identification}\HCode{<span class="inline-javascript" id="javascript\
1202 </htXimera>
```

2.9 SageMath support

Load SageTeX if it exists.

```
1203 <*classXimera>
1204 \IfFileExists{sagetex.sty}{\RequirePackage{sagetex}}{}
1205 </classXimera>
```

`sageCell (env.)` Create an interactive SageMath widget.

```
1206 <*classXimera>
1207 \DefineVerbatimEnvironment{sageCell}{Verbatim}{numbers=left,frame=lines,label=SAGE,labelposi
1208 </classXimera>
```

```

1209 <*htXimera>
1210 \renewenvironment{sageCell}{\NoFonts}{\EndNoFonts}
1211 \ScriptEnv{sageCell}{\ifvmode \IgnorePar\fi \EndP\HCode{<div class="sage"><script type="text/
1212 </htXimera>

sageOutput (env.)    Execute SageMath code and output the result.
1213 <*classXimera>
1214 \DefineVerbatimEnvironment{sageOutput}{Verbatim}{numbers=left,frame=lines,label=SAGE-Output,
1215 </classXimera>

1216 <*htXimera>
1217 \renewenvironment{sageOutput}{\NoFonts}{\EndNoFonts}
1218 \ScriptEnv{sageOutput}{\ifvmode \IgnorePar\fi \EndP\HCode{<div class="sageOutput"><script ty
1219 </htXimera>

sageSilent (env.)    Execute SageMath code without outputting the result.
1220 <*htXimera>
1221 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
1222 \ifdefined\sagesilent
1223 \renewenvironment{sagesilent}{\NoFonts}{\EndNoFonts}
1224 \fi
1225 \ScriptEnv{sagesilent}{\ifvmode \IgnorePar\fi \EndP\HCode{<script type="text/sagemath">}\Htm
1226 </htXimera>

```

2.10 Answerables

2.10.1 Answers

```

\answer A math answer
1227 <*classXimera>
1228
1229 \ifdefined\HCode
1230 \newcommand{\recordvariable}[1]{
1231 \else
1232 \newwrite\idfile
1233 \immediate\openout\idfile=\jobname.ids
1234 \newcommand{\recordvariable}[1]{\ifthenelse{\equal{#1}{}}{\immediate\write\idfile{var #1};}
1235 \fi

Determines if answer is shown in handout mode. when given=true, show answer in
handout mode, show answer in “given box” outside handout mode. When given=false,
do not show answer in handout mode, show answer outside handout mode
1236 \define@key{answer}{given}[true]{\def\ans@given{#1}}

Used for setting numeric answer tolerance for online student input.
1237 \define@key{answer}{tolerance}{\def\ans@tol{#1}}

Used to run dynamic js code on student provided answers. Note: currently pdf outputs
the validator code itself.
1238 \define@key{answer}{validator}{}

Used for assigning a js ID to answer for dynamic code (eg validators).
1239 \define@key{answer}{id}{\def\ans@id{#1}}

Used to set anticipated input format; eg “string”.
1240 \define@key{answer}{format}{}

Used to hide the answer input box on the web.
1241 \define@key{answer}{onlinenoinput}[false]{}

Used to add a ‘show answer’ button to the answer blank.
1242 \define@key{answer}{onlineshowanswerbutton}[false]{}

Set default values for \answer command key=value pairs. Default values are given = false.
1243 \setkeys{answer}{id=,given=false,onlinenoinput=false,onlineshowanswerbutton=false}

```

Basic code for `\answer`.

```

1244
1245 % Options for handout
1246 \newcommand{\answerFormatLength}{2cm}
1247
1248 \newcommand{\answerFormatDots}[1]{\ldots\ldots}
1249 \newcommand{\answerFormatLine}[1]{\protect\rule{\answerFormatLength}{0.4pt}}
1250 \newcommand{\answerFormatFlexibleLine}[1]{\protect\rule{\widthof{${#1}$}*2}{0.4pt}}
1251 \newcommand{\answerFormatFlexibleBox}[1]{\fbox{\scalebox{2}{\phantom{${#1}$}}}}
1252
1253 % options for default (i.e with answers filled in)
1254 \newcommand{\answerFormatPlain}[1]{\ensuremath{#1}}
1255 \newcommand{\answerFormatBlue}[1]{\color{blue}\ensuremath{#1}}
1256 \newcommand{\answerFormatBoxed}[1]{\fbox{\ensuremath{#1}}}
1257 \newcommand{\answerFormatBoxedGiven}[1]{\underset{\scriptstyle\mathrm{given}}{\fbox{\ensuremath{#1}}}}
1258
1259 % defaults for handout and default mode, and for \answer[given]
1260 \let\handoutAnswerFormat\answerFormatDots
1261 \let\defaultAnswerFormat\answerFormatBlue
1262 \let\givenAnswerFormat\answerFormatBoxedGiven
1263
1264 \newcommand{\answer}[2][\%
1265   \ifmode%
1266     \setkeys{answer}{#1}%
1267     \recordvariable{\ans@id}
1268     \ifthenelse{\boolean{\ans@given}}{
1269       {% Start then statement
1270         \ifhandout
1271           #2
1272         \else
1273           \givenAnswerFormat{#2} %% in case the argument helps formatting
1274         \fi
1275       }% End then statement
1276     }{% Start else statement
1277       \ifhandout
1278         \handoutAnswerFormat{#2} %% in case the argument helps formatting
1279       \else% show answer in box outside handout mode
1280         \defaultAnswerFormat{#2} %% in case the argument helps formatting
1281       \fi
1282     }% End else statement
1283   \else%
1284     \GenericError{\space\space\space\space}% Throw an error based on... something? -- Jason
1285     {Attempt to use \@backslashchar answer outside of math mode}
1286     {See https://github.com/ximeraProject/ximeraLatex for explanation.}
1287     {Need to use either inline or display math.}%
1288   \fi
1289 }
1290 \endclassXimera

```

On the HTML side, `\answer` emits spans—but it is usually just handled directly by MathJax.

```

1291 \beginXimera
1292 \renewcommand{\answer}[2][false]{\HCode{<span class="answer_respondable">#2\HCode{</span>}}
1293
1294 \def\validator[#1]{\stepcounter{identification}\HCode{<div class="validator" id="validator@a
1295 \def\endvalidator{\HCode{</div>}}
1296
1297 \endXimera

```

2.10.2 Multiple choice and the like

`multipleChoice (env.)` Multiple choice

```

1298 \beginXimera

```

1299 % Jim: Originally this was \renewcommand{\theenumi}{\$(\mathrm{\alph{enumi}})\$}
 1300 % but that breaks tex4ht because mathmode can only be processed by mathjax.
 1301 % so now I made this just italicized.

2.10.3 Options

1302 \define@key{choice}{value}[]{\def\choice@value{#1}}

This flags the answer as the correct answer

1303 \define@boolkey{choice}{correct}[true]{\def\choice@correct{#1}}

Use an ID to refer to the choice.

1304 \define@key{multipleChoice}{id}{\def\mc@id{#1}}

\otherchoice outputs the item if correct and nothing if incorrect.

1305 \define@key{otherchoice}{value}[]{\def\otherchoice@value{#1}}

1306 \define@boolkey{otherchoice}{correct}[true]{\def\otherchoice@correct{#1}}

Default key choices for multiple choice options. Default for choice pairs. Default: answers without the option "correct=true" is "incorrect".

1307 \setkeys{choice}{correct=false,value=}

Defaults for multipleChoice pairs. Default to no id? – Jason

1308 \setkeys{multipleChoice}{id=}

Defaults for otherchoice pairs. Default "otherchoice" to behave like "choice" for error checking.

1309 \setkeys{otherchoice}{correct=false,value=}

1310 </classXimera>

2.10.4 Choices

\choice Like \item but for choice environments. choice command denotes a possible answer choice for the multiple choice question.

1311 <*classXimera>

1312 \newcommand{\choice}[2][]{%

1313 \setkeys{choice}{#1}%

1314 \item{#2}

1315 \ifthenelse{\boolean{\choice@correct}}{

1316 {% Begin then result

1317 \ifhandout% if it's a handout do nothing.

1318 \else% otherwise place a checkmark when you select the "correct choice"... maybe? -- Jason

1319 \,\checkmark\,\setkeys{choice}{correct=false}

1320 \fi

1321 }% End then result

1322 }% Begin/End else result.

1323 }

1324

1325 %Define an expandable version of choice Not really meant to be used outside this package (use

1326 % Is there a reason we can't just always use this as default? -- Jason

1327 \newcommand{\choiceEXP}[2][]{%

1328 \expandafter\setkeys\expandafter{choice}{#1}%

1329 \item{#2}

1330 \ifthenelse{\boolean{\choice@correct}}{

1331 {% Begin then result

1332 \ifhandout

1333 \else

1334 \,\checkmark\,\setkeys{choice}{correct=false}

1335 \fi

1336 }% End then result

1337 }% Begin/End else result.

1338 } %% note all the {} are needed in case the choice has [] in it.

1339

1340 % \otherchoice is the \choice used in wordChoice command.

1341 \newcommand{\otherchoice}[2][]{%

1342 \ignorespaces%

1343 \setkeys{otherchoice}{#1}%

1344 \ifthenelse{\boolean{\otherchoice@correct}}{

```

1345 {% Start then result
1346 #2\ignorespaces\setkeys{otherchoice}{correct=false}\ignorespaces%
1347 }% End then result
1348 {}% Start/End else result
1349 \ignorespaces%
1350 }%
1351 \newcommand{\inlinechoice}[2][{}]{%
1352 \setkeys{choice}{#1}%
1353 \iffirstinlinechoice
1354 (\hspace{-.25em}
1355 \firstinlinechoicetrue
1356 \else
1357 /
1358 \fi
1359 #2
1360 \ifthenelse{\boolean{choice@correct}}{%
1361 {% Start then result
1362 \ifhandout\else\checkmark\ignorespaces\setkeys{choice}{correct=false}\ignorespaces\fi%
1363 }% End then result
1364 {}% Start/End else result
1365 \hspace{-.25em}\ignorespaces%
1366 }
1367
1368 \end{classXimera}

```

On the HTML side, `\choice` emits `<span>s`.

```

1369 (*htXimera)
1370 \newcounter{choiceId}
1371 \renewcommand{\choice}[2][{}]{%
1372 \setkeys{choice}{correct=false}%
1373 \setkeys{choice}{#1}%
1374 \stepcounter{choiceId}\IgnorePar%
1375 \HCode{<span class="choice }%
1376 \ifthenelse{\boolean{choice@correct}}{\HCode{correct}}{}%
1377 \HCode{" }
1378 \ifthenelse{\equal{\choice@value}{}}{\HCode{data-value="\choice@value" }}%
1379 \HCode{id="choice\arabic{choiceId}">}%
1380 #2\HCode{</span>}}
1381 \let\inlinechoice\choice
1382 \end{htXimera}

```

2.10.5 Environment(s)

`multipleChoice (env.)` The environment `multipleChoice@` is for internal use only. Wrap `\choices` in a `multipleChoice` environment to make a multiple choice question.

```

1383 (*classXimera)
1384 \newenvironment{multipleChoice}[1][{}]{%
1385 {% Environment Start Code
1386 \setkeys{multipleChoice}{#1}%
1387 \recordvariable{mc@id}%
1388 \begin{trivlist}
1389 \item[\hspace{\labelsep}\small\bfseries \GetTranslation{Multiple Choice}:]\hfil
1390 \begin{enumerate}
1391 }% Note this means that \item has to be the first line after \begin{multipleChoice}.
1392 {% Environment End Code
1393 \end{enumerate}
1394 \end{trivlist}
1395 }
1396
1397 %multipleChoice@ is for internal use only! (used in wordChoice)
1398 %this is simply a wrapper for the sole showing (other)choice.
1399 \newenvironment{multipleChoice@}[1][{}]{%
1400 \end{classXimera}

```

On the web, you might also expect these to be "problem environments" but they aren't – they're responsables. You might expect a `\setcounter{choiceId}{0}` here — that would be wrong, because then the generated IDs would no longer be unique.

```

1401 <*htXimera>
1402 \renewenvironment{multipleChoice}[1][
1403 {\setkeys{multipleChoice}{#1}%
1404 \stepcounter{identification}\ifvmode \IgnorePar\fi \EndP\HCode{<div class="multiple-choice"
1405 \ifthenelse{\equal{\mc@id}{}}{\}\HCode{data-id="\mc@id" }}%
1406 \HCode{id="problem\arabic{identification}" titletext=" \GetTranslation{Multiple Choice}">}%
1407 }\HCode{</div>}\IgnoreIndent}
1408 \ConfigureEnv{multipleChoice}{\}\{\}\{\}
1409 </htXimera>

```

2.11 Word choice

`\wordChoice` An in-line version of `multipleChoice`: uses `enumitem` package note, it is coded as a single line to avoid unwanted spaces in "given" mode.

```

1410 <*classXimera>
1411 \newcommand{\wordChoice}[1]{%
1412 \let\choicetemp\choice% Assign a "choicetemp" command to duplicate choice.
1413 \ifwordchoicegiven% If wordchoice option is on, we need to juggle around some definitions.
1414 \let\choice\otherchoice%
1415 %\begin{multipleChoice@}% -unnecessary (REMOVE THIS LINE IF THE YEAR IS 2019 or Beyond)
1416 #1
1417 %\end{multipleChoice@}% -unnecessary (REMOVE THIS LINE IF THE YEAR IS 2019 or Beyond)
1418 \else% If it isn't the regular "choice" command should work.
1419 \let\choice\inlinechoice%
1420 \begin{multipleChoice@}%
1421 #1%
1422 \end{multipleChoice@}%
1423 \fi%
1424 \let\choice\choicetemp% Now that choicetmp has been manipulated to what we want, replace choi
1425 }%
1426
1427
1428 </classXimera>

```

This is actually just word choice

```

1429 <*htXimera>
1430 \renewenvironment{multipleChoice@}{\refstepcounter{problem}}{\}%
1431 \ConfigureEnv{multipleChoice@}{\stepcounter{identification}\IgnorePar\HCode{<span class="word
1432 </htXimera>

```

2.12 Select all

`selectAll (env.)` A multiple-multiple choice question

```

1433 <*classXimera>
1434 \newenvironment{selectAll}[1][
1435 {\begin{trivlist}\item[\hspace \labelsep\small\bfseries \GetTranslation{Select All Correct Ans
1436 {\end{enumerate}}\end{trivlist}}
1437 </classXimera>

```

In the future we need this to (optionally) be displayed in the problem, while the actual code lives in the solution. Here is how this could be implemented: Like the `title/maketitle` commands, the `multiple-choice` could be stored in `\themultiplechoice`, flip a boolean, and execute `\makemultiplechoice` at the `\end` of the problem. We should also make a command called `\showchoices` that will show choices in the handout.

On the web, `selectAll` is handled just like `multipleChoice`.

```

1438 <*htXimera>
1439 \renewenvironment{selectAll}{\refstepcounter{problem}}{\}%
1440 \ConfigureEnv{selectAll}{\stepcounter{identification}\ifvmode \IgnorePar\fi \EndP\HCode{<div
1441 </htXimera>

```

2.12.1 Free response

freeResponse (*env.*) A freeform input box.

```

1442 <*classXimera>
1443 \newboolean{given} %% required for freeResponse
1444 \setboolean{given}{true} %% could be replaced by a key=value pair later if needed
1445
1446 \ifhandout
1447 \newenvironment{freeResponse}[1][false]%
1448 {%
1449 \def\givenatend{\boolean{#1}}
1450 \ifthenelse{\boolean{#1}}
1451 {% Begin then result
1452 \begin{trivlist}
1453 \item
1454 }% End then result
1455 {% Begin else result
1456 \setbox0\vbox\bgroup
1457 }% End else result
1458 % {}% Don't think this is doing anything? -- Jason
1459 }
1460 {%
1461 \ifthenelse{\givenatend}
1462 {% Begin then result
1463 \end{trivlist}
1464 }% End then result
1465 {% Begin else result
1466 \egroup
1467 }% End else result
1468 % {}% Don't think this is doing anything? -- Jason
1469 }
1470 \else
1471 \newenvironment{freeResponse}[1][false]%
1472 {% Environment Beginning Code
1473 \ifthenelse{\boolean{#1}}%% Could probably change this with just putting the (given) in t
1474 {% Begin then result
1475 \begin{trivlist}
1476 \item[\hspace{1em}\labelsep\bfseries \GetTranslation{Free Response (Given)}:\hspace{2ex}]
1477 }% End then result
1478 {% Begin else result
1479 \begin{trivlist}
1480 \item[\hspace{1em}\labelsep\bfseries \GetTranslation{Free Response}:\hspace{2ex}]
1481 }% End else result
1482 }
1483 {% Environment Ending Code
1484 \end{trivlist}
1485 }
1486 \fi
1487
1488 </classXimera>
1489 <*htXimera>
1490
1491 \renewenvironment{freeResponse}{\refstepcounter{problem}}{}%
1492 \ConfigureEnv{freeResponse}{\stepcounter{identification}\ifvmode \IgnorePar\fi \EndP\HCode{<
1493
1494 </htXimera>

```

2.12.2 Feedback

feedback (*env.*) An initially hidden environment that uncovers itself at an appropriate time. New Validator rewrite code added by Jason Nowell. Original code provided by Jim Fowler. Validator is an environment designed to run a custom check on answers (usually) using javascript code.

Define a placeholder command for validator and feedback.

```

1495 <classXimera>
1496 \newcommand{\PH@Command}{}

Validator should take an argument and detokenize it and display it at the start of the
environment. The original Validator environment had everything framed in an mbox;
presumably to make the text look a bit nicer, although this seems redundant with texttt.
It shouldn't cause any harm so I have left it in for now.

1497 \newenvironment{validator}[1][]{
1498   \def\PH@Command{\#1}% Use PH@Command to hold the content and be a target for "\expandafter"
1499   \mbox{\texttt{\detokenize\expandafter{\PH@Command}}}% Now expand PH@Command once and then d
1500 }{}

```

First, if it's a handout, we want feedback to eat everything and then disappear entirely. So we do this:

```

1501 \ifhandout%
1502 \newenvironment{feedback}
1503   {%
1504     \setbox0\vbox\bgroup
1505     }
1506   {%
1507     \egroup
1508     }

```

If this isn't a handout, then we want to display the Feedback by using a label, positioned and formatted as a `\item` in a trivlist. It is important that we also detokenize the content of the optional argument, as it is likely to contain javascript or other code that latex won't be able to make sense of.

```

1509 \else
1510 \newenvironment{feedback}[1][attempt]{
1511
1512 \edef\PH@Command{\GetTranslation{\#1}}% Use PH@Command to hold the content and be a target for
1513
1514 \begin{trivlist}% Begin the trivlist to use formatting of the "Feedback" label.
1515 \item[\hspace{1em}\labelsep\small\slshape\bfseries \GetTranslation{Feedback}]% Format the "Feedback
1516 \ifonlineTF{% If the feedback is on a pdf, we don't need to detokenize - which messes with t
1517 (\texttt{\expandafter\detokenize\expandafter{\PH@Command}})}% Keep the online version the sam
1518 {(\expandafter\texttt{\PH@Command}}):% No need for detokenize in the pdf version
1519 \hspace{2ex}}\small\slshape% Insert some space before the actual feedback given.
1520 }{
1521 \end{trivlist}
1522 }
1523
1524 \fi
1525 </classXimera>

```

Feedback environments take an optional parameter (which describes when the feedback is to be provided)

```

1526 <htXimera>
1527 \def\feedback{\@ifnextchar{\@feedbackcode}{\@feedbackattempt}}
1528 \def\@feedbackattempt{\@feedbackcode[attempt]}
1529 \def\@feedbackcode{\#1}{\stepcounter{identification}%
1530 \ifvmode \IgnorePar\fi \EndP%
1531 \ifthenelse{\equal{\#1}{attempt}}{\HCode{<div class="feedback" data-feedback="attempt" id="fe
1532 {\ifthenelse{\equal{\#1}{correct}}{\HCode{<div class="feedback" data-feedback="correct" id="f
1533 {\HCode{<div class="feedback" data-feedback="script" id="feedback\arabic{identification}" ti
1534 \def\endfeedback{\HCode{</div>}\IgnoreIndent}
1535 </htXimera>

```

2.12.3 Ungraded activities

`ungraded (env.)` The `ungraded` environment is used to record that certain parts of activities should not be worth points. For example, if you want to use a `multipleChoice` as a survey question,

you can place it inside an `ungraded` environment. On the L^AT_EX side, the `ungraded` environment does nothing.

```
1536 \classXimera
1537 \newenvironment{ungraded}{}{}
1538 \endclassXimera
```

But on the html side, `ungraded` wraps the activities in a `div` in order to assign some weight to them for grading.

```
1539 \htXimera
1540 \renewenvironment{ungraded}{%
1541 \ifmode \IgnorePar\fi \EndP\HCode{<div class="ungraded">}\IgnoreIndent%
1542 }{
1543 \ifmode \IgnorePar\fi \EndP\HCode{</div>}\IgnoreIndent%
1544 }
1545 \endhtXimera
```

2.13 Support for the web

2.13.1 MathJax support

When using mathjax, dump all the `\newcommands` to a `.jax` file.

First, create the `.jax` file. Redefine newcommand appropriately.

```
1546 \classXimera
1547 %% Pre-202412: .jax file written in non-\HCode, and in a next run inserted by ximera.cfg in
1548 %% Post-202501: .mjax file written only in \HCode, and in luaxake post-processing inserted in
1549 %% ( used luaxake rather than sed ...)
1550 \newwrite\myfile
1551 \ifdefined\HCode
1552 \immediate\openout\myfile=\jobname.xmjax
1553
1554 %% From |only.dtx| we must also create |prompt| on the MathJax side.
1555 \immediate\write\myfile{\unexpanded{\newenvironment}{prompt}}{}{}
1556
1557 %% Write all newcommands to .xmjax file, that will be included in the .html via luaxake
1558 \let\@oldargdef\@argdef
1559 \long\def\@argdef#1[#2]#3{%
1560 \immediate\write\myfile{\unexpanded{\newcommand}{\unexpanded{#1}}[\unexpanded{#2}]{\unexpanded{#3}}%
1561 \@oldargdef#1[#2]#3}%
1562 }
1563
1564 %% Same for \DeclareMathOperator
1565 \let\@oldDeclareMathOperator\DeclareMathOperator
1566 \renewcommand{\DeclareMathOperator}[2]{\@oldDeclareMathOperator{#1}{#2}\immediate\write\myfile{\unexpanded{\newcommand}{\unexpanded{#1}}[\unexpanded{#2}]{\unexpanded{#3}}%
1567 }
1568 \fi
1569
1570
1571 \endclassXimera
```

Include the jax'ed newcommands (pre-202412 versions)

```
1572 \cfigXimera
1573
1574 % 202501: removed sed-manipulation of .jax file; see luaxake now
1575
1576 \Configure{BVerbatimInput}{}{}{}
1577
1578 \Configure{verbatiminput}{}{}{}
1579
1580 % Instead of a nonbreaking space, use a standard space
1581 \makeatletter
1582 \def\FV@Space{\space}
1583 \makeatother
1584
1585 % Include the (problem-?) .ids in a text/javascript script right at the beginning of the body
```

```

1586 \Configure{BODY}{%
1587 \HCode{<body>\Hnewline}%
1588 \Tg<div class="preamble">%
1589 %% 202501: removed .jax inclusion (see luaxake)
1590
1591 %% Include the .ids file
1592 \IfFileExists{\jobname.ids}{\HCode{<script type="text/javascript">\Hnewline}%
1593 \BVerbatimInput{\jobname.ids}%
1594 \HCode{</script>\Hnewline}%
1595 }{}
1596 \Tg</div>%
1597 }{%
1598 \ifvmode\IgnorePar\fi\EndP\HCode{</body>\Hnewline}%
1599 }
1600
1601 % 202501: removed 'prevent spaces as in "\begin {align}": this is done in luaxake now
1602
1603 % This is a fix for the LAODE book, which uses matlabEquation as if it were an equation
1604 \ScriptEnv{matlabEquation}{\ifvmode \IgnorePar\fi \EndP\HCode{<script type="math/tex; mode=d
1605
1606 \</cfgXimera>

```

2.13.2 Semantic HTML

`\textbf` Using `\textbf` emits a `<strong>` tag.

```

1607 \<cfgXimera>
1608 \Configure{textbf}{\ifvmode\ShowPar\fi\HCode{<strong>}}{\HCode{</strong>}}
1609 \</cfgXimera>

```

`\textit` Using `\textit` or similar emits an `<em>` tag.

```

1610 \<cfgXimera>
1611 \Configure{textit}{\ifvmode\ShowPar\fi\HCode{<em>}}{\HCode{</em>}}
1612 \Configure{emph}{\ifvmode\ShowPar\fi\HCode{<em>}}{\HCode{</em>}}
1613 \</cfgXimera>

```

`\texttt` Using `\texttt` emits a `<code>` tag.

```

1614 \<cfgXimera>
1615 \Configure{texttt}{\ifvmode\ShowPar\fi\HCode{<code>}}{\HCode{</code>}}
1616 \</cfgXimera>

```

2.14 Tools

2.14.1 Suppress

`suppress (env.)` The suppress environment is a good way to suppress output without commenting it. This way we can avoid many of the places we use `environ` package and this should also avoid most of the verbatim conflicts. This is code adapted from `syntonly.sty`.

```

1617 \<classXimera>
1618 \font\dummyft@=dummy \relax
1619 \def\suppress{%
1620   \begingroup\par
1621   \parskip\z@
1622   \offinterlineskip
1623   \baselineskip=\z@skip
1624   \lineskip=\z@skip
1625   \lineskiplimit=\maxdimen
1626   \dummyft@
1627   \count@\sixt@@n
1628   \loop\ifnum\count@ >\z@
1629     \advance\count@\m@ne
1630     \textfont\count@\dummyft@
1631     \scriptfont\count@\dummyft@
1632     \scriptscriptfont\count@\dummyft@
1633   \repeat

```

```

1634 \let\selectfont\relax
1635 \let\mathversion\@gobble
1636 \let\getanddefine@fonts\@gobbletwo
1637 \tracinglostchars\z@
1638 \frenchspacing
1639 \hbadness\@M}
1640 \def\endsuppress{\par\endgroup}
1641 \endclassXimera

```

2.14.2 The End

It seems that some of the files need to conclude with something or another.

```

1642 \endhtXimera}
1643 \Hinput{ximera}
1644 \endhtXimera}

1645 \endhtXourse}
1646 \Hinput{xourse}
1647 \endhtXourse}

1648 \endcfgXimera}
1649 \begin{document}
1650 \EndPreamble
1651 \endcfgXimera}

```

3 xourse.cls

```

1652 \classXourse}

```

notoc The default behavior of the class is to provide a table of contents listing all activities in the course. This option will suppress this table of contents.

```

1653 \newif\ifnotoc
1654 \notocfalse
1655 \DeclareOption{notoc}{\notoctrue}

```

nonewpage The default behavior of the class is to start each activity on a new page. This option will start activities without making a new page.

```

1656 \newif\ifnonewpage
1657 \nonewpagefalse
1658 \DeclareOption{nonewpage}{\nonewpagetrue}

1659 \DeclareOption*{\PassOptionsToClass{\CurrentOption}{ximera}}
1660 \ProcessOptions\relax
1661 \LoadClass{ximera}
1662 % \begin{macrocode}
1663 \endclassXourse}

```

3.1 Activities

The core of the **xourse** system. It works by redefining the **document** environment, thus making the **\begin** and **\end{document}** of the subfile ‘transparent’ to the inclusion. The redefinition of **\documentclass** is analogous, just having a required and an optional arguments which mean nothing to **\subfile**.

```

1664 \classXourse}
1665 \newcommand{\skip@preamble}{%
1666   \let\document\relax\let\enddocument\relax%
1667   \newenvironment{document}{\let\input\otherinput}{}%
1668   \renewcommand{\documentclass}[2][subfiles]{}}

```

Note that the new command **\subfile** calls for **\skip@preamble** *within a group*. The changes to **document** and **\documentclass** are undone after the inclusion of the subfile.

Numbering starts a page too soon without this:

```

1669 \let\otherinput\input

```

```

Store usual \maketitle as \othermaketitle
1670 \let\othermaketitle\maketitle
\maketitle In a xourse file, \maketitle is redefined to give course packet title page and toc.
1671 \renewcommand{\maketitle}{%
1672 \pagestyle{empty}
1673 \begin{center}
1674 ~\ \ %puts space at top of page to move title down.
1675 \vskip .25\textheight
1676 \hrulefill\
1677 \vskip 1em
1678 \bfseries{\Huge \@title} \
1679 \hrulefill\
1680 \vskip 3em
1681 {\Large \@author}
1682 \vskip 2em
1683 {\large \@date}
1684 \end{center}
1685 \clearpage

When notoc option is used, we do not include a table of contents. Otherwise we include
a table of contents in every course packet.
1686 \ifnotoc
1687 \else
1688 \tableofcontents\clearpage
1689 \clearpage
1690 \fi

Switch to main pagestyle, just like a document with documentclass ximera.
1691 \pagestyle{main}

Renew maketitle to usual definition.
1692 \let\maketitle\othermaketitle

And we finish with our redefinition of \maketitle.
1693 }
1694 \relax
1695 \end{classXourse}

```

3.1.1 Regular activities

\activity Documents included with **\activity** will be included in the body of the xourse document. Any **\input** commands within included ximera documents will be ignored. Any **\usepackage** commands within included ximera documents will cause an error. Overlapping **\newcommand** definitions within multiple ximera documents included simultaneously will cause an error. The **\activity** command inputs the file name provided without **\documentclass**, without **\begin{document}**/**\end{document}** and without any inputs in the preamble of the included file.

```

1696 \begin{classXourse}
1697 \ifnonepage
1698 \newcommand{\activity}[2][\%]{%
1699 \setkeys{activity}{#1}
1700 \renewcommand{\input}[1]{%
1701 \begin{group}\skip@preamble\otherinput{#2}\end{group}\par\vspace{\topsep}
1702 \let\input\otherinput}
1703 \else
1704 \newcommand{\activity}[2][\%]{%
1705 \setkeys{activity}{#1}
1706 \renewcommand{\input}[1]{%
1707 \begin{group}\skip@preamble\otherinput{#2}\end{group}\clearpage
1708 \let\input\otherinput}
1709 \fi
1710 \relax
1711 \end{classXourse}

```

```

1712 <*htXourse>
1713 \renewcommand\activity[2][]{%
1714 \ifvmode \IgnorePar\fi \EndP\HCode{<a class="activity card \activitystyle" href="#2" data-op
1715 }
1716 </htXourse>

```

When running xake, we can just ignore activities

```

1717 <*classXourse>
1718 \ifxake
1719 \renewcommand\activity[2][]{%
1720 \fi
1721 </classXourse>

```

3.1.2 Practice activities

`\practice` Like `\activity` but not expecting a title.

```

1722 <*classXourse>
1723 \ifhandout
1724 \newcommand{\practice}[2][]{%
1725 \setkeys{practice}{#1}%!!!!
1726 \renewcommand{\input}[1]{%
1727 \begingroup\skip@preamble\otherinput{#2}\endgroup
1728 \let\input\otherinput}
1729 \else
1730 \newcommand{\practice}[2][]{\texttt{\detokenize{#2}}}% gives file name for practice
1731 \setkeys{practice}{#1}%!!!!
1732 \renewcommand{\input}[1]{%
1733 \begingroup\skip@preamble\otherinput{#2}\endgroup
1734 \let\input\otherinput}
1735 \fi
1736 \relax
1737 </classXourse>

```

The practice environment does nothing, but will eventually produce exercises at the end of an activity

```

1738 <*classXourse>
1739 \ifxake
1740 \renewcommand\practice[2][]{%
1741 \fi
1742 </classXourse>

```

I suppose it is reasonable for practice cards to NOT have an `activitystyle`, since the `activitystyle` is basically PRACTICE.

```

1743 <*htXourse>
1744 \renewcommand\practice[2][]{%
1745 \ifvmode\IgnorePar\fi\EndP%
1746 \HCode{<a class="activity card practice" href="#2" data-options="#1">#2</a>}%
1747 \IgnoreIndent%
1748 }
1749 </htXourse>

```

3.2 Sectioning

Makes the table of contents look a bit better. This can be redefined in the preamble if you do not like the appearance. The name of a section inside an activity.

```

1750 <*classXourse>
1751 \renewcommand*\l@section{\@dottedtocline{1}{1.5em}{4.2em}}
1752 </classXourse>

```

`\subsection` The name of a subsection inside an activity.

```

1753 <*classXourse>
1754 \renewcommand*\l@subsection{\@dottedtocline{2}{3.8em}{4.2em}}
1755 </classXourse>

```

`\part` Xourse files can have parts. The name of a large part of a xourse.

```

1756 <*htXourse>
1757 \newcounter{ximera@part}
1758 \setcounter{ximera@part}{0}
1759 \renewcommand\part[1]{%
1760 \stepcounter{ximera@part}%
1761 \ifvmode \IgnorePar\fi \EndP%
1762 %\HCode{<h1 id="part\arabic{ximera@part}" class="card part">}#1\HCode{</h1>}}% makes cards di
1763 \HCode{<h1 id="part\arabic{ximera@part}" class="card part">}#1</h1>}}%
1764 \IgnoreIndent%
1765 }
1766 </htXourse>

```

`\paragraph` Paragraph commands emit spans. A small heading.

```

1767 <*cfgXimera>
1768 \renewcommand{\paragraph}[1]{%
1769   \HCode{<span class="paragraphHead">}}%
1770   #1%
1771   \HCode{</span>}\par\IgnorePar}
1772 </cfgXimera>

```

`\subparagraph` An even smaller heading.

```

1773 <*cfgXimera>
1774 \renewcommand{\subparagraph}[1]{%
1775   \HCode{<span class="subparagraphHead">}}%
1776   #1%
1777   \HCode{</span>}\par\IgnorePar}
1778 </cfgXimera>

```

3.3 Grading by points

`graded (env.)` The `graded` environment does nothing in latex, but in html, it wraps the activities in a div in order to assign some weight to them for grading.

```

1779 <*classXourse>
1780 \newenvironment{graded}[1]{\{}
1781 </classXourse>

```

So indeed this environment in html wraps the activities in a div in order to assign some number of points to them.

```

1782 <*htXourse>
1783 \renewenvironment{graded}[1]{%
1784 \ifvmode \IgnorePar\fi \EndP\HCode{<div class="graded" data-weight="#1">}\IgnoreIndent%
1785 }{
1786 \ifvmode \IgnorePar\fi \EndP\HCode{</div>}\IgnoreIndent%
1787 }
1788 </htXourse>

```

3.4 Logos

`\logo` A logo for the xourse.

```

1789 <*classXourse>
1790 \newcommand*{\logo}[1]{%
1791   \ifx\@onlypreamble\@notprerr
1792     \ClassError{xourse}{logo can only be used in the preamble}
1793     {Move your logo command to the preamble}
1794   \else %
1795     \IfFileExists{#1}%
1796     {\gdef\xourse@logo{#1}}%
1797     {\ClassError{xourse}{logo file does not exist}
1798      {To use logo, make sure that the referenced image file exists}}%
1799   \fi%
1800 }
1801
1802 </classXourse>

```

The xourse logo is an `og:image` in the opengraph taxonomy.

```
1803 <*htXourse>
1804 \Configure{@HEAD}{%
1805   \HCode{<meta name="og:image" content="}%
1806   \ifdefined\xourse@logo%
1807     \xourse@logo%
1808   \fi%
1809   \HCode{" />\Hnewline}}%
1810 </htXourse>
```